Jiyu Hu

201 N Goodwin Ave | (217) 904-5744
Urbana, IL, 61801 | jiyuhu2@illinois.edu

Education

University of Illinois Urbana-Champaign Aug. 2023 – Present
Ph.D. in Computer Science *Urbana, IL*

Carnegie Mellon University – School of Computer Science Aug. 2021 – May 2023
Master of Computational Data Science *Pittsburgh, PA*

University of Illinois Urbana-Champaign Aug. 2017 – May 2021
Bachelor of Science in Computer Engineering *Urbana, IL*

Publications

* = Equal Contribution

Kiran Hombal, Jiyu Hu, Marcos K. Aguilera, Aishwarya Ganesan, and Ramnatthan Alagappan. Disk-Based LSMs: An Unexpectedly Good Index for Partly Coherent CXL Memory. In *Proceedings of the ACM SIGOPS 32th Symposium on Operating Systems Principles*, **SOSP '26**, New York, NY, USA, 2026. Association for Computing Machinery. (To Appear)

Jiyu Hu, Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan. MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory. In *20th USENIX Symposium on Operating Systems Design and Implementation*, **OSDI '26**. USENIX Association, 2026. (To Appear)

Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, and Aishwarya Ganesan. LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications. *ACM Transactions on Computer Systems*, August 2025

Shreesha G. Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, and Ramnatthan Alagappan. Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering. In *19th USENIX Symposium on Operating Systems Design and Implementation*, **OSDI '25**, pages 465–481. USENIX Association, 2025

Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, and Aishwarya Ganesan. LazyLog: A New Shared Log Abstraction for Low-Latency Applications. In *Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles*, **SOSP '24**, page 296–312, New York, NY, USA, 2024. Association for Computing Machinery

Jiyu Hu, Jack Kosaian, and K. V. Rashmi. Rethinking Erasure-Coding Libraries in the Age of Optimized Machine Learning. In *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems*, **HotStorage '24**, page 23–30, New York, NY, USA, 2024. Association for Computing Machinery

Rui Yang, Jiangran Wang, Jiyu Hu, Shichu Zhu, Yifei Li, and Indranil Gupta. Medley: A Membership Service for IoT Networks. *IEEE Transactions on Network and Service Management*, 19(3):2492–2505, 2022

Xueda Shen*, Jiyu Hu*, Yunqi Zhang, and Ian C. Quinn. B2-Coupon: Efficient and Non-intrusive Mobile Coupon Distribution using Dual Bloom Filter. In *2020 IEEE/ACM Symposium on Edge Computing (SEC)*, pages 358–363, 2020

Presentations

Rethinking Erasure Coding Libraries in the Age of Optimized Machine Learning Jul. 2024
HotStorage '24 *Santa Clara, CA*

B^2 **Coupon** Nov. 2020
ACM SEC (workshop talk)

Research Experience

Prism Jun. 2025 – Jun. 2026
DASSL, University of Illinois Urbana-Champaign *Urbana, IL*

- Investigate efficient index design on CXL memory under the assumption of partial hardware coherence.

MEGALON Apr. 2024 – Dec. 2025
DASSL, University of Illinois Urbana-Champaign *Urbana, IL*

- Investigate the design of efficient sharing of CXL memory under the assumption of partial hardware coherence.

SpecLog Apr. 2024 – Dec. 2024
DASSL, University of Illinois Urbana-Champaign *Urbana, IL*

- Design and implement a new shared log abstraction that significantly decreases e2e latency by speculation.

LazyLog Jan. 2024 – Jul. 2024
DASSL, University of Illinois Urbana-Champaign *Urbana, IL*

- Design and implement a new shared log abstraction that significantly decreases append latency from state-of-the-art shared log implementations by delaying ordering the appended log entries.

TVM-EC Jan. 2022 – Jul. 2024
TheSys Lab, Carnegie Mellon University *Pittsburgh, PA*

- Propose a new way of implementing high-performance erasure-coding libraries via machine learning libraries, reducing the effort to design and maintain erasure coding libraries.

Medley Jan. 2020 – Jul. 2021
Distributed Protocols Research Group, University of Illinois Urbana-Champaign *Urbana, IL*

- Develope and evaluate a new IoT failure detection protocol that is aware of the spatial locality of physical nodes so as to decrease the overall communication overhead in an unstable network environment.

B^2**-Coupon** Nov. 2019 – Nov. 2020
Prof. Dong Xuan's Research Group, The Ohio State University *Columbus, OH*

- Design a better coupon distribution protocol for mobile devices.

Awards & Scholarships

Student Travel Grant
OSDI '25 & OSDI '26 *2025, 2026*

Best Paper Award (LazyLog)
SOSP '24 *2024*

Senior Design Instructor's Award (The Best Senior Design Award)
University of Illinois Urbana-Champaign *2021*

Bradley A. Simons Memorial Scholarship
University of Illinois Urbana-Champaign *2019*

Dean's List
University of Illinois Urbana-Champaign *2017, 2018, 2019, 2020, 2021*

Services

CAIS ’26, Artifact Evaluation Committee

EuroSys ’26, Shadow PC

SOSP ’25, Artifact Evaluation Committee

FAST ’26, Artifact Evaluation Committee

OSDI ’24, ATC ’24, Artifact Evaluation Committee

Teaching

CS 498 Cloud Computing Applications — TA
University of Illinois Urbana-Champaign — *2026*

Industry Experience

GSL, Microsoft Corporation — May 2026 – Aug. 2026
Research Intern (Supervisor: Rathijit Sen)
- Work on disaggregated memory for database systems.

USM, Exadata, Oracle America, Inc. — May 2022 – Aug. 2022
Software Intern
- Worked on the kernel log aggregation and analysis framework of Oracle distributed database.

CUDA Math Library, NVIDIA Corporation — June 2020 – Aug. 2020
Software Intern
- Analyzed the floating point error propagation in cuBLAS GEMM kernel.